

# Tax AI Infrastructure

The plumbing behind the next three years of tax — and why the org chart, not the model, is the bottleneck.

A working thesis, grounded in data from in-house tax-technology teams.

## THE QUESTION

**In 2028, what part of the tax function will still be done **the way it is today?****

My answer: less than you think — but not for the reason the vendors are selling you.

WHERE THIS COMES FROM

# Not vibes — a benchmark of real tax-tech functions

## Anonymous

In-house tax & tax-technology teams report how they actually run: headcount, automation, architecture, AI stage.

## Vendor-neutral

A non-profit, community-built dataset. No product to sell, no "magic quadrant" to defend.

## Operational

9 dimensions — from build-vs-buy to data flow to regulatory response time — not survey-of-CFOs fluff.

Throughout this talk, the numbers come from what teams report about themselves — so we're calibrating to reality, not roadmaps.

# Three shifts define the next 2-3 years

## SHIFT 01

### Software → Infrastructure

We stop buying "tax apps" and start assembling a stack: data, grounding, models, agents, governance.

## SHIFT 02

### Batch → Real-time

E-invoicing and continuous controls force tax from period-end batch to event-driven, always-on.

## SHIFT 03

### Headcount → Oversight

The job shifts from *doing* the work to *supervising* agents that do it. Trust becomes the product.

The winners won't have the best model. They'll have the best **infrastructure** around an ordinary one.

# 01

## PART ONE

# Where tax technology actually is today

Before we talk about the future, let's be honest about the floor we're starting from.

# The tax stack is still mostly manual and batch

## ERP + XLS

Still the dominant "tax data architecture" — no dedicated layer

## Batch ETL

Data moves period-end, not in real time. Manual transfers are common.

## Weeks

Typical time to respond to a regulatory change

You cannot bolt autonomous agents onto a spreadsheet you reconcile by hand once a quarter.

# Most teams are at "exploration," not production

## What's happening

- **Exploration & PoC** dominate the GenAI adoption stage.
- Use cases cluster in **research & drafting** — low-risk, human-checked.
- Few teams report **production** or enterprise-wide deployment.

## Why it stalls

- No **data foundation** to ground the model in your facts.
- No **evaluation** harness → can't prove it's right → can't ship.
- Ownership is split across **Tax, Finance Systems, and IT**.

The gap between a demo and production **is the infrastructure**.  
That's the whole game.

# Three forcing functions are converging

## 1 · Capability

Frontier models now do multi-step reasoning, tool use, and long-context document work well enough for real tax tasks.

## 2 · Regulation

E-invoicing & continuous transaction controls (ViDA, national mandates) make tax data *real-time and machine-readable* by law.

## 3 · Pressure

Pillar Two, more jurisdictions, flat headcount. The work scales; the team doesn't. Something has to give.

Capability makes it **possible**. Regulation makes it **necessary**. Pressure makes it **urgent**.



# 02

PART TWO

## The Tax AI Infrastructure stack

Six layers. You build them bottom-up. Skip one and the layer above it collapses.

# From raw transactions to supervised autonomy

6

## Governance & Evals

Accuracy tests, audit trail, explainability, human sign-off

5

## Orchestration

Tools, ERP connectors, MCP, workflow & approvals

4

## Agents

Research · provision · compliance · controversy

3

## Models

Frontier foundation models + domain prompting/tuning

2

## Knowledge & Grounding

Tax law, positions, prior filings — retrievable & cited

1

## Data Foundation

Clean, real-time, lineage-tracked transaction & entity data

Read it bottom-up. Most failed AI pilots tried to start at layer 4.

# If the data isn't real-time and trusted, stop here

- A **tax data layer** over the ERP — not 40 spreadsheets and a SharePoint.
- **Event-driven ingestion** so transactions arrive as they happen.
- **Lineage** — every number traces back to a source you can defend in an audit.
- Entity, jurisdiction, and product master data that's actually **maintained**.

1 Data Foundation

2 Knowledge

3 Models

4 Agents

This is unglamorous, expensive, and the single highest-ROI thing you can do for AI.

# The model knows tax in general. It must know **your** tax.

## What lives here

- Statutes, regs, rates — by jurisdiction, with effective dates.
- Your **tax positions**, memos, and prior-year filings.
- Rulings, e-invoicing schemas, and a jurisdiction knowledge graph.

## Why it matters

- Retrieval grounds answers in **citable sources** — not memory.
- "Show me the authority" becomes a built-in, not a hope.
- This is what turns a chatbot into a **defensible** tool.

No grounding, no citations. No citations, no auditor will ever accept the output.

# Models are the commodity. Don't fall in love with one.

- Frontier foundation models, swapped as they improve — assume **portability**.
- Most "tuning" you need is **prompting + grounding**, not training your own model.
- Right-size: a cheap fast model for extraction, a strong model for reasoning.
- Treat the model as a **replaceable part**, not the architecture.

↓ 10×

Rough order-of-magnitude drop in cost-per-task each generation. Plan for it.

The teams that hard-wired one vendor's model into everything will be re-platforming in 18 months.

# From "ask a question" to "do the task"

## Research agent

Reads the law + your positions, drafts a memo with citations and an explicit confidence flag.

## Compliance agent

Prepares returns from the data layer, reconciles, flags anomalies for human review.

## Provision agent

Pulls trial balance, computes the provision, drafts the workpapers and the narrative.

## Controversy agent

Assembles audit responses from prior filings, source data, and the document trail.

An agent is a model **plus tools, plus a goal, plus guardrails** — it acts, it doesn't just chat.

# Agents are only as useful as what they can reach

- **Connectors** to the ERP, the data layer, the filing portals, the tax engine.
- **MCP & tool standards** so any agent can call any system through one interface.
- **Workflow & approvals** — where a human signs, where an agent may proceed.
- **Observability** — logs of every action, ready for review and audit.

## The integration tax

Benchmark data shows most teams are still on **vendor tax engines + ERP**, glued with batch jobs. Orchestration is where AI either plugs into reality — or stays a sandbox demo.

This is the layer that quietly decides whether your pilot ever touches a real return.

# This is the moat — and the part everyone skips

## Evaluate

A test set of known-answer tax cases. You measure accuracy *before* you trust output.  
Regression-test every model swap.

## Audit

Every answer carries its sources, inputs, and the reasoning trail.  
Reproducible months later.

## Control

Confidence thresholds route low-certainty items to humans. Clear sign-off and segregation of duties.

In tax, **being right is not enough — you must be able to prove you were right.** Governance is that proof.



# 03

## PART THREE

# Three stages of AI evolution in tax

AI-Native Foundation → Continuous Learning → Proactive Management.  
The order is not optional.

# How AI evolves through the tax function

STAGE 1 • 2026



## AI-Native Foundation

- Data, grounding & AI built in — not bolted on
- Copilots assist; humans own every step
- You're laying track, not running trains yet

STAGE 2 • 2027



## AI Continuous Learning

- System learns from your corrections & outcomes
- Adapts to new law, rates & positions automatically
- Accuracy compounds; agents run tasks for review

STAGE 3 • 2028



## AI Proactive Management

- AI spots issues & acts *before* you ask
- Near-autonomous — **human approval retained**
- Tax team supervises; AI manages the operation

Each stage requires the one before it to be real. You earn the next stage with evidence — jurisdiction by jurisdiction.

# This isn't a forecast — coding is already running the playbook

Software engineering is the same three stages, ~2-3 years ahead. It's the leading indicator for what reaches tax next.

STAGE 1 • DONE ✓

## AI-Native Foundation

Writing code *with* AI is now the default. Tools like Claude Code and AI-first editors made the transformation — it's complete, not coming.

STAGE 2 • IN PROGRESS ↻

## Continuous Learning

The strategic prize is now **data access** — the usage signal (accepts, edits, fixes) that lets the tools learn. That's why AI-native coding tools command a premium. Ongoing, not finished.

STAGE 3 • EMERGING ⚙

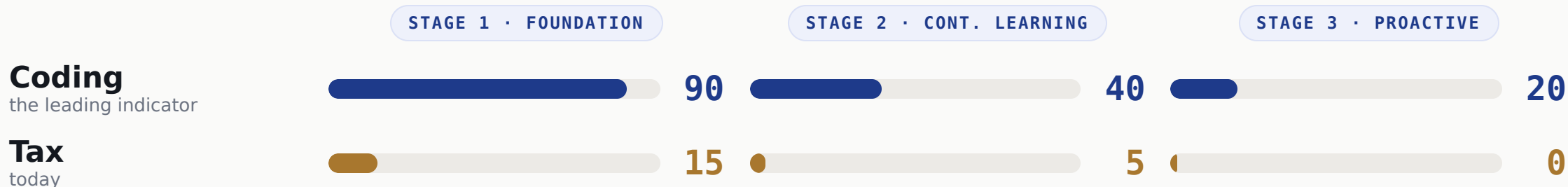
## Proactive Management

Early signs — agents that proactively refactor, fix, open a PR. Real, but **not autonomous**: a human still approves the merge.

Tax is at Stage 1 today. Coding proves the sequence holds — and that the human-approval gate survives all the way through.

## THE GAP, QUANTIFIED

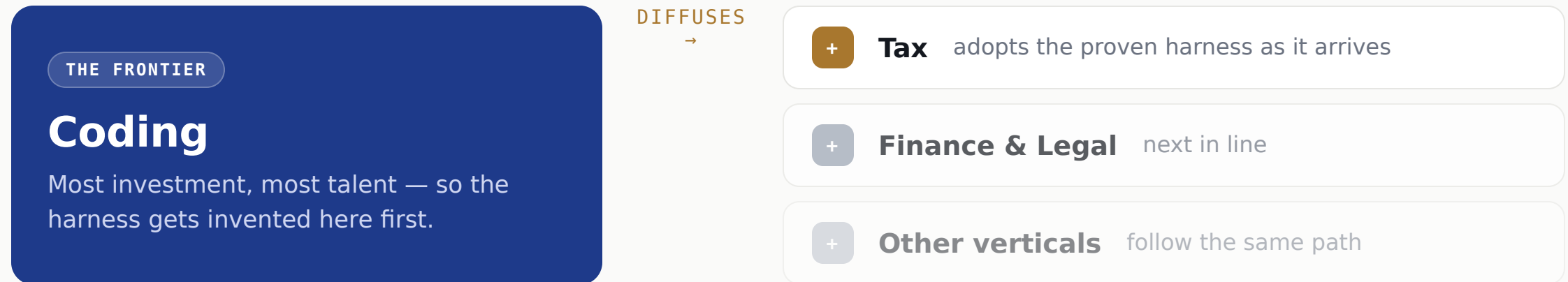
# If we scored the three stages, 0-100



Same curve, ~2-3 years apart. Coding has finished Stage 1 and is mid-Stage 2 — **which is exactly the map tax gets to read ahead.**

# The harness for Stage 2 & 3 doesn't exist yet — anywhere

Continuous Learning and Proactive Management need a **harness** — the scaffolding that makes models learn and act reliably. Nobody has built it. It's being figured out in coding, then it diffuses.



Even most AI startups are still at the **Foundation** stage. So you can't buy Stage 2 or 3 today — but you *can* build your foundation, and be ready the moment the harness diffuses.

# Build the AI-Native Foundation across the whole tax lifecycle

You don't modernize one function in isolation. You build **one** AI-native foundation — and every stage of the tax lifecycle draws from it.

## Tax Planning

Scenario modeling and structuring research, grounded in your own positions and the law.

## Compliance & Reporting

High-volume, rules-based work on real-time data — the natural first mover.

## Tax Audit

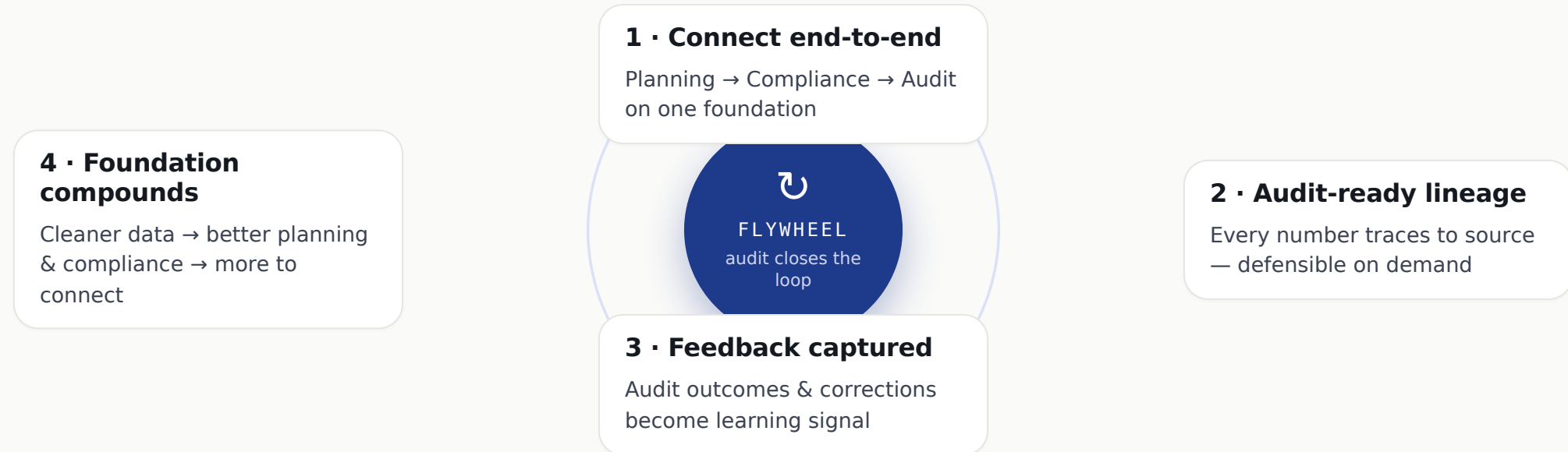
Controversy & defense built on the same lineage, sources, and document trail.

**1 One AI-Native Foundation** Shared data · grounding · governance — under all three

Foundation first. Build it once, build it well — then Continuous Learning and Proactive Management have somewhere to stand.

# Connect the lifecycle end-to-end — a flywheel kicks in

Not a cost you pay before the payoff. Connect Planning → Compliance → Audit on one foundation and each loop compounds — **audit closes the loop.**



Each turn makes the data cleaner, the audit position stronger, the foundation smarter — **the exact feedback engine Stage 2 will run on.**

# What an AI-native tax foundation actually requires

"Foundation" is not one thing you buy. It's eight infrastructure disciplines — each hard on its own, and only valuable when integrated.

## Data platform / lakehouse

Snowflake · Databricks ·  
BigQuery — a real tax data lake

## Streaming & pipelines

Kafka · Flink · CDC — event-  
driven, not batch ETL

## Integration & connectors

ERP · tax engines · APIs · MCP  
tool standards

## Semantic & master data

Entity / jurisdiction models ·  
catalog · lineage

## Retrieval / knowledge

Vector store · embeddings ·  
document parsing (RAG)

## AI / ML platform

Model serving · GPU compute ·  
agent orchestration

## Governance & security

Access control · PII / privacy  
· audit logging

## Eval & observability

Test harness · monitoring ·  
end-to-end tracing

Each box is its own engineering team in most companies. **The foundation is all eight, integrated** — that's the bar.



# Why almost no tax function can stand this up

## Messy, fragmented data

Tax data is scattered across ERPs, billing, supply chain, and spreadsheets — in many jurisdictions, with no clean source of truth.

## Real-time is a different sport

Most teams live in batch ETL. Streaming, CDC, and event-driven infra are a specialized, expensive skill set they don't have.

## Doubly-rare talent

You need people fluent in tax *and* platform/data/AI engineering. Each is scarce; the overlap barely exists.

## It's a multi-year platform

Not a SKU you buy — a cross-cutting build that spans eight disciplines and years, not a quarter.

## Split ownership

Tax vs. Finance Systems vs. IT. No single owner, no clear mandate, no budget line that fits.

## Tax is a cost center

Funding hyperscale-grade infrastructure for a function seen as overhead is a hard sell to any CFO.

This is why "just add AI" stalls. The blocker isn't the model — it's **standing up eight infra disciplines a tax team was never built**

# Why we were uniquely positioned to build it

## Hyperscale data infra exists

One of the largest data platforms in the world is already here. Tax builds *on* it — we don't stand up a lakehouse from scratch.

## Real-time by default

The company runs event-driven and streaming end-to-end. Tax data can be real-time native, not a batch afterthought.

## World-class engineering

Deep bench of platform, data, and ML engineers — and a culture where tax-tech is embedded with real software teams.

## AI platform & compute in-house

Frontier models, GPU scale, and mature ML tooling already operating — not a procurement exercise.

## Scale justifies the build

Enormous transaction volume across many jurisdictions makes the ROI on a real foundation obvious.

## Engineering-owned tax tech

The org model lets tax move like a product team — own the stack, ship, and iterate.

We didn't have to build the infra *and* the harness. The infra was already here — so we could put it to work for tax.

# What gets automated first — and last

## Automates early

- High-volume, rules-heavy **indirect tax / compliance**
- Data extraction & reconciliation
- First-draft research & documentation
- E-invoicing & transactional reporting

## Automates last

- Novel **planning & structuring** judgment
- Controversy strategy & negotiation
- Anything with high ambiguity & high stakes
- Final sign-off & accountability — always human

Volume and structure get automated. **Judgment and accountability stay human** — and become more valuable.

# 04

## PART FOUR

# What it means for your team

Build-vs-buy, org design, talent, and the trust problem.

# The line moves — buy the model, own the stack

## Buy

Foundation models, tax engines, e-invoicing connectors, eval tooling. Commodities — don't build these.

## Own

Your data foundation, your grounded knowledge, your evals, your governance. This is your moat.

## Assemble

Orchestration glues bought parts to owned data. Increasingly low-code — but a real engineering discipline.

Benchmark reality: teams are split build/buy/hybrid — but the durable answer is **buy the commodity, own the context.**

# The team composition inverts

- **Fewer preparers, more reviewers** — humans supervise agent output.
- A new role: the **tax engineer** — fluent in tax *and* data/AI.
- Ownership consolidates: AI forces Tax, Finance Systems & IT to **actually co-own** the stack.
- Skill mix shifts from spreadsheet ops to **data engineering & prompt/eval design**.

## The benchmark signal

Teams already split tax-tech FTEs across frontend, backend, **data engineering**, and DevOps — and locate tax-tech in Tax, Finance Systems, or IT. The orgs that resolve that ownership question first will move fastest.

The scarce hire of 2028 isn't a tax expert or an engineer. It's the person who is **credibly both**.

# What can go wrong — and how the stack answers it

## Hallucinated authority

Model invents a citation. → **Grounding + evals** force every claim back to a real source.

## Silent errors at scale

One bad rule, ten thousand returns. → **Confidence thresholds + sampling + audit trail.**

## Regulator acceptance

"Explain this number." → **Reproducible lineage** from output to source data.

## Data leakage / privacy

Sensitive data to a model. → **Governance, access control, deployment choices.**

Every risk maps to a layer. That's not a coincidence — **the stack is the risk-management strategy.**

# A maturity model you can place yourself on

- 3 Proactive Management** AI acts ahead of the work; near-autonomous; human approval retained
- 2 Continuous Learning** System learns from feedback & adapts to law; agents run tasks for review
- 1 AI-Native Foundation** Copilots in daily use; grounded, with a real data foundation in place
- 0 Exploratory** PoCs, ungrounded chatbots, no evals — where most teams sit today

Most of the room is at **Level 0**. The next three years is climbing the three stages — **in order**.



# Five moves that survive any model

## 1 · Fix the data foundation

It's the bottleneck and the highest-ROI work. Start with the data e-invoicing already gives you.

## 2 · Build an eval set

50 known-answer tax cases. Now you can measure any tool — and prove it works.

## 3 · Resolve ownership

Decide where tax-tech lives. Get Tax, Finance Systems & IT co-owning one stack.

## 4 · Pick one workflow

One high-volume, rules-based process. Ground it, instrument it, ship a copilot.

**5 · Stay model-agnostic.** Architect so you can swap the model in an afternoon, not a quarter.

**Models will commoditize.**  
**Infrastructure compounds.**  
**Build the stack, in order, and  
earn autonomy with  
evidence.**

Software → Infrastructure

Batch → Real-time

Headcount → Oversight

# This thesis gets sharper with your numbers in it

## **Anonymous**

Benchmark your automation, team structure, and AI stage against peers — no names attached.

## **Non-profit**

Community-built, free, vendor-neutral. The dataset belongs to the field, not a SKU.

## **Equal access**

Contribute once, see the whole picture. No premium tier, no gatekeeping.

If you run an in-house tax-tech function, you have a data point this field needs.

TRY IT LIVE

# Contribute & compare

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before the talk and drop the  
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driven. Your data calibrates the next  
three years.

THANK YOU

# Questions?

Let's talk about what's broken in your stack — that's where the next three years actually happen.

[contact@taxbenchmark.ai](mailto:contact@taxbenchmark.ai) · [taxbenchmark.ai](https://taxbenchmark.ai)